

So $Ax=b$ has a solution when $3b_1+b_2=0$ $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$
 $Ax=b$ has no solution when $3b_1+b_2 \neq 0$. $\square$

16. Repeat Exercise 15: $A = \begin{bmatrix} 1 & -3 & -4 \\ -3 & 2 & 6 \\ 5 & -1 & -8 \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Solution: By Theorem 3, the equation $Ax=b$ has the same solution as the ^{system of} equations corresponding to the augmented matrix $[a_1 \ a_2 \ a_3 \ b]$.

$$\begin{bmatrix} 1 & -3 & -4 & b_1 \\ -3 & 2 & 6 & b_2 \\ 5 & -1 & -8 & b_3 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 3 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} - 5 \times \text{Row 1}} \begin{bmatrix} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & 3b_1+b_2 \\ 5 & -1 & -8 & b_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & 3b_1+b_2 \\ 0 & 14 & 12 & b_3-5b_1 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \\ \text{Row 3} + 2 \times \text{Row 2}}]{\text{Row 3} =} \begin{bmatrix} 1 & -3 & -4 & b_1 \\ 0 & -7 & -6 & 3b_1+b_2 \\ 0 & 0 & 0 & b_1+2b_2+b_3 \end{bmatrix}$$

Verification of whether the final matrix is in echelon form

1. ~~Row~~ If $b_1+2b_2+b_3 \neq 0$, there are no zero rows in the matrix.
 $\therefore$ The condition that all nonzero rows must be above all the zero rows is vacuously True.
 If $b_1+2b_2+b_3 = 0$, Row 3 is a zero row. Row 1 and Row 2 are nonzero rows. $\therefore$ The condition that all nonzero rows must be above all the zero rows is True.

2. Row 1 Leading entry = 1 in Column 1
 Row 2 Leading entry = -7 in Column 2
 Row 3 possible Leading entry = $b_1+2b_2+b_3$ in Column 4
 So, the condition that the leading entry of each row is in a column to the right of the columns of the leading entries of all rows above it is True.